

# sloopDX Cheat Sheet

DX7 FM synth + FM drums for the M-VAVE FM-1  
Version 3.1 · dx7.designburgapps.com · github.com/zvenson/dxsloop

## THE FOUR TRACKS

1 2 3 = DX7 synths (8 voices shared) · 4 = drums (6 voices of their own). Every track has its own pattern.

## BASICS

|                      |                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>ALGORITHM</b>     | select the track (on every page)                                                                                         |
| <b>PRESETS</b>       | sound: 01–20 factory, 21–52 your bank, then user presets · drums: the kit                                                |
| <b>SELECT</b>        | tempo                                                                                                                    |
| <b>KNOB 1–4</b>      | HOME: swing · level · steps · pan                                                                                        |
| <b>PLAY</b>          | start / stop all tracks                                                                                                  |
| <b>REC</b>           | playing: record now · stopped: armed, the first note starts · empty project: play freely, REC on the “1” closes the loop |
| <b>hold REC</b>      | clear the track (a ring fills, ~2 s)                                                                                     |
| <b>OCT– / OCT+</b>   | octave (both = 0) · drums, held: ghost / hard hits                                                                       |
| <b>HOME</b>          | TRACKS · hold: menu (colour, low cut, lights, USB audio, about)                                                          |
| <b>EDIT + OCT–/+</b> | undo / redo                                                                                                              |

## LAYERS: HOLD A BUTTON

|             |                                                                          |                                   |
|-------------|--------------------------------------------------------------------------|-----------------------------------|
| <b>Hold</b> | White keys                                                               | <b>KNOB 1 · 2 · 3 · 4</b>         |
| <b>FX</b>   | punch-in effect                                                          | filter · dust · duck              |
| <b>EDIT</b> | erase that sound / note                                                  | shift · length ×2½ · transp.      |
| <b>ARP</b>  | note repeat (roll)                                                       | rate                              |
| <b>SEQ</b>  | steps · <b>OCT–/+</b> pages 1–8 · step held: K4 tune, PRESETS decay lock | sound/note · div · swing · length |
| <b>SCL</b>  | the key of the song                                                      | chord · scale · keys · transp.    |
| <b>GLO</b>  | 1–4 mute · 5–8 solo · 16 tap                                             | level of tracks 1–4               |
| <b>SAVE</b> | 1–4 play section A–D · 5–8 store · 13 loop/song · 14 song rec            |                                   |

Lock a layer: hold its button + tap **HOME**. Tap instead of hold: its pages.

**OMNI** (ARP page, MODE OMNI): black keys = chords F C G · Dm Am · Em G7 E7 · D7 Bb · A7, white keys = strings, swipe. MODE FLW: a track's pattern follows the chord.

## EDIT A DX7 VOICE: TAP **EDIT**

|                  |                                                                              |
|------------------|------------------------------------------------------------------------------|
| <b>SELECT</b>    | choose the row                                                               |
| <b>ALGORITHM</b> | change the value (the first turn on “Algorithm” only draws it)               |
| <b>EDIT</b>      | open a group / run an action (tap twice)                                     |
| <b>HOME</b>      | one level back                                                               |
| <b>PRESETS</b>   | straight to the next / previous operator                                     |
| <b>KNOB 1–4</b>  | CUT · RESO (the low-pass) · REL · FDBK; value at the bottom, dials top right |
| <b>SAVE</b>      | store the bank (a dot at the bottom = not stored)                            |

**List:** Voice · Bank 1–8 · Algorithm 1–32 · Feedback · Osc Sync · Transpose · **OP1–OP6** · Pitch Env · LFO · Name · Copy To · Init Voice · Store · More Pages ·

**Operator:** its envelope drawn (KNOB 1–4: Rate 1–4, on a Level row Level 1–4; your note runs along it) · On · Solo · Output Level · Coarse (ratio) · Fine · Detune · Osc Mode · Rate 1–4 · Level 1–4 · Key Velocity · Amp Mod Sens · Break Point · Left/Right Depth · Left/Right Curve · Rate Scaling

Changing a factory voice copies it into the first free user slot first (“COPIED TO U01”). In the algorithm drawing: carriers cyan, modulators light blue.

## SHAPE THE SOUND

|                        |                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EDIT pages</b>      | KNOB 1–4: VOICE · BRITE · ATK · DEC, then REL · FDBK · CUT · RESO                                                                                                                                                            |
| <b>BRITE</b>           | the modulators' level: dull ↔ bright                                                                                                                                                                                         |
| <b>ATK · DEC · REL</b> | slower / longer (0 = as programmed)                                                                                                                                                                                          |
| <b>FDBK</b>            | feedback: sine → saw                                                                                                                                                                                                         |
| <b>CUT · RESO</b>      | a low-pass behind the voice (CUT 127 = open, out of the way)                                                                                                                                                                 |
| <b>ENV page</b>        | ATK · DEC · REL of the carriers, SUS lowers the sustain (127 = as programmed)                                                                                                                                                |
| <b>ENV/LFO DEST</b>    | FLT = the cutoff, SHP = brightness, PIT = pitch, AMP = volume                                                                                                                                                                |
| <b>FX pages</b>        | SENDS (the four as dials: DIST · CHO · DLY · REV), then one per effect, the send first: DIST drive · tone · type · mix — CHORUS send · rate · depth · mix — DELAY send · time · fdbk · mix — REVERB send · size · damp · pre |

## DRUMS: TRACK 4, WHITE KEYS

|               |              |             |             |            |              |             |               |
|---------------|--------------|-------------|-------------|------------|--------------|-------------|---------------|
| F3<br>Kick    | G3<br>Kick 2 | A3<br>Snare | B3<br>Clap  | C4<br>Hat  | D4<br>Open   | E4<br>Pedal | F4<br>Rim     |
| G4<br>Snare 2 | A4<br>Tom L  | B4<br>Tom H | C5<br>Crash | D5<br>Ride | E5<br>Shaker | F5<br>Conga | G5<br>Cowbell |

**Kits** (PRESETS): 1 DX KIT · 2 808 FM · 3 ELECTRO · 4 METAL · 5 **MY KIT**. A black key plays the sound left of it. **SEQ**: grid ↔ kit. **Kit page** KNOB 1–4: kit · level · **drive** · **comp**. **EDIT**: the **lane** (the sound last played): TUNE · DECAY · SWEEP · BRIGHT, **PRESETS**: NOISE · LEVEL · PAN · CHOKE, then REV · **dice** kit / sound (turn twice). **SAVE** 2×: MY KIT.

## YOUR OWN DX7 BANKS (.SYX)

- Chrome / Edge: [dx7.designburgapps.com](https://dx7.designburgapps.com) → Editor → **Connect**.
- Tab **Library**: pick “Bank in use” 1–8, drop the .syx (32 voices, 4104 bytes).
- On the FM-1: PRESETS 21–52, or EDIT → Voice. Switch banks: EDIT → Bank.

8 banks × 32 = 256 voices in memory. Original DX7 ROMs: [yamahablackboxes.com](https://yamahablackboxes.com) · collections: [bobbyblues.recup.ch](https://bobbyblues.recup.ch)

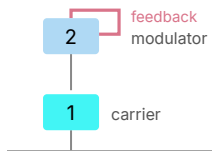
## SAVE & RESCUE

|                |                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------|
| <b>Project</b> | tap <b>SAVE</b> → PROJECT: slot, turn twice = save / load (4 slots, autosave)                           |
| <b>Sound</b>   | <b>SAVE</b> → USER: 32 user presets                                                                     |
| <b>Backup</b>  | editor → Projects → Backup (projects, presets, DX7 bank)                                                |
| <b>Update</b>  | <a href="https://dx7.designburgapps.com">dx7.designburgapps.com</a> → Install (USB straight in, no hub) |
| <b>Go back</b> | M-UPGRADE + the original firmware · emergency: FM-1-transporter                                         |



## OPERATORS

A DX7 voice is made of **six operators**. Each is a sine oscillator with its own **envelope** and **level**. An operator is either a **carrier** (you hear it) or a **modulator** (it bends the wave of the operator below it: the more level, the more harmonics). That is FM: the brightness comes from the modulators. sloopDX adds a low-pass behind each voice (CUT, RESO) for the classic synth sweep.



Operator 2 modulates operator 1. Light blue = modulator, cyan = carrier, the pink loop = **feedback** (an operator modulating itself: from sine to saw to noise).

## ALGORITHMS 1-32

The **algorithm** says how the six are wired: stacks (2 → 1, deep and bright), pairs (5: three electric-piano-like pairs), one modulator on several carriers (22), or all six as carriers (32: an organ of sines). On the FM-1: EDIT → *Algorithm*, the first turn draws it, the next ones change it.

## REACHING THE OPERATORS

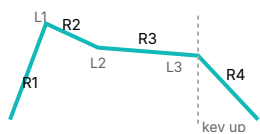
1. Synth track (1-3): tap **EDIT**. The voice list opens.
2. **SELECT** to **OP1** ... **OP6**, tap **EDIT**: that operator's rows.
3. **PRESETS**: the next / previous operator, on the same row (compare Rate 1 of all six quickly).
4. **ALGORITHM** changes the value, **HOME** goes back, **SAVE** stores.
5. Row **On**: switch the operator off to hear what it does (not stored).

In the web editor (tab **Voice**) all six are on one page, with their envelopes drawn.

## FREQUENCY: COARSE, FINE, DETUNE

|                 |                                                                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Coarse</b>   | the ratio to the note: 0.50, 1, 2, 3 ... 31. Whole ratios on a modulator = harmonic (organ, brass, bass); 1.41, 3.5 ... = inharmonic (bells, metal) |
| <b>Fine</b>     | between the ratios (1.00 → 1.99)                                                                                                                    |
| <b>Detune</b>   | -7...+7: two carriers slightly apart beat (chorus, e-piano shimmer)                                                                                 |
| <b>Osc Mode</b> | FIXED: a frequency regardless of the key (1, 10, 100, 1000 Hz ranges): noises, drums, clicks                                                        |

## ENVELOPES: 4 RATES, 4 LEVELS



Rates are **speeds** (99 = instant, low = slow), levels are **targets**. R1→L1 attack, R2→L2 and R3→L3 decay to the sustain L3 (held while the key is down), R4→L4 release. A carrier's envelope = volume, a modulator's = brightness over time.

## TOUCH AND THE KEYBOARD

|                                  |                                                                       |
|----------------------------------|-----------------------------------------------------------------------|
| <b>Key Velocity</b>              | 0-7: harder = louder (carrier) or brighter (modulator)                |
| <b>Rate Scaling</b>              | 0-7: high notes decay faster, as a piano                              |
| <b>Break Point, Depth, Curve</b> | level up or down the keyboard from a key: thinner bass, softer treble |
| <b>Amp Mod Sens</b>              | how much the LFO's AMD reaches this operator (tremolo)                |

## PITCH ENVELOPE AND LFO

**Pitch Env**: 4 rates, 4 levels for the pitch of all operators (level 50 = no shift): a blip on the attack, a slow drop. **LFO**: wave (triangle, saw, square, sine, S&H), speed, delay, **PMD** (vibrato) with Pitch Mod Sens, **AMD** (tremolo) with Amp Mod Sens, key sync.

## FM DRUMS: THE LANE PAGE

Each drum is a DX7 voice too: a **body** (a carrier, swept down at the hit), modulators for **tone**, and **noise operators** (drums only: noise instead of a sine, its frequency the colour). On the kit page tap **EDIT** for the sound you last played: **TUNE** the note, **DECAY** every falling rate, **SWEEP** the pitch drop, **BRIGHT** the modulators, **NOISE** the noise, then **LEVEL**, **PAN**, **CHOKE** and the **REV** send. Locks put a TUNE / DECAY on one hit.

## RECIPES: START FROM 20 INIT VOICE

|                  |                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Bass</b>      | Alg 1 or 5 · OP2 Coarse 1, Level 70-85, fast decay (R2 ~70, L2 0) · OP1 carrier · Feedback 4-6 for growl                         |
| <b>Sub / 808</b> | OP1 alone at Coarse 0.50: a pure sine an octave down · 808: Pitch Env L4 66, R1 72 (it drops into the note), L3 0 (it dies away) |
| <b>E-piano</b>   | Alg 5 · carriers Coarse 1 with Detune ±2 · one modulator Coarse 14 with a short decay (the tine), Key Velocity 5-7               |
| <b>Bell</b>      | Alg 5 · modulator Coarse 3.5 (Fine) Level 70 · carrier long R3/R4 · L3 0 (it rings out)                                          |
| <b>Brass</b>     | Alg 22 · modulator envelope with a slower R1 than the carriers (the tone opens), Feedback 7, a little Pitch Env attack           |
| <b>Organ</b>     | Alg 32 · six carriers at Coarse 0.5, 1, 2, 3, 4, 6 (the drawbars), Levels set the mix                                            |
| <b>Pad</b>       | slow R1 (30-45) on the carriers, long R4, two carriers detuned ±3, LFO PMD a little                                              |

Then: **Copy To** a slot, **Name** it, **SAVE**. Factory voices are copied into a user slot automatically on the first change.